MEDOVIR – PACKAGE INSERT

1. NAME OF THE MEDICINAL PRODUCT
Medovir, 400mg, tablets

2. QUALITATIVE AND QUANTITATIVE COMPOSITION
Each Medovir 400 mg tablet contains 400mg of aciclovir.

Excipient with known effect: lactose monohydrate.
Each Medovir 400mg tablet contains 90.5mg lactose.
For the full list of excipients, see section 6.1

3. PHARMACEUTICAL FORM
Medovir 400mg Tablet is light pink, round, scored on one side with diameter 12.7mm

4. CLINICAL PARTICULARS
4.1. Therapeutic indications
Medovir is indicated for:
- The treatment of herpes simplex virus infections of the skin and mucous membranes including initial and recurrent genital herpes
- The suppression of recurrent herpes simplex infections in immunocompetent patients
- The prophylaxis of herpes simplex infections in immunocompromised patients
- The treatment of varicella (chickenpox) and herpes zoster (shingles) infections.

4.2. Posology and method of administration
Posology
Adults:
Treatment of *herpes simplex* infection: 200mg five times a day at approximately four hour intervals, omitting the night time dose. Treatment is usually for five days, in the case of severe initial infections this may need to be extended. In patients with impaired gut absorption, or severely immunocompromised, dose may be doubled to 400mg (or the intravenous form may be used).
Therapy should be initiated as early as possible, preferably during the prodromal period or when lesions first appear in the case of recurrent episodes.

Suppression of *herpes simplex* infections in immunocompetent patients: 200mg four times a day at approximately six hour intervals. Dose of 400mg two times a day at approximately twelve hour intervals is convenient for many patients. In some cases, dosage titration down to 200mg three times a day at eight hour intervals, or 200mg two times a day at twelve hour intervals, may be effective.
Cases of breakthrough infection on total doses of 800mg a day may occur in some patients.
Every six to twelve months, therapy should be periodically interrupted to observe any possible changes in the disease.

Prophylaxis of *herpes simplex* infection in immunocompromised patients: 200mg four times a day at approximately six hour intervals. In patients with impaired gut absorption, or severely immunocompromised, dose may be doubled to 400mg (or the intravenous form may be used).
Length of treatment for prophylaxis is determined by the period of risk.

Treatment of *varicella* and *herpes zoster* infections: 800mg five times a day at approximately four hour intervals, omitting the night time dose. Treatment should last seven days. In patients with impaired gut absorption, or severely immunocompromised, use of intravenous administration should be considered.

Herpes zoster treatment gives better results if therapy initiation is as soon as possible after rash appearance, treatment of chickenpox in immunocompromised patients should be initiated within 24 hours of rash appearance.

Children:
Treatment of, and prophylaxis in immunocompromised, *Herpes simplex* infection: children two years of age or older, as the adult dose. Children under two years of age, half the adult dose it is recommended.
Treatment of *Varicella* infections: treatment should last five days.

Age six years and over	:	800mg four times a day
Age two to five years	:	400mg four times a day
Age under two years	:	200mg four times a day

Dosing may be calculated more accurately as 20mg/kg bodyweight four times a day. Total daily dosage must not exceed 800mg.
No information is available on suppression of *Herpes simplex* infection or the treatment of *Herpes zoster* infection in immunocompetent children.

Elderly: Together with creatinine clearance, total aciclovir body clearance declines. Adequate hydration of such patients must be maintained. Attention should be paid to the need for dosage reduction, especially in such patients with renal impairment.

Renal impairment: Recommended doses for *Herpes simplex* infection will not lead to accumulation of aciclovir above infusion levels in patients with impaired renal function. In severe renal impairment, creatinine clearance <10ml/min, 200mg two times a day at approximately twelve hour intervals is recommended.
In treatment of *Varicella* and *Herpes zoster* infection, the following dosage adjustments are recommended:
- moderate renal impairment (creatinine clearance 10ml - 25ml/min): 800mg three times a day at approximately eight hour intervals.
- severe renal impairment (creatinine clearance <10ml/min): 800mg two times a day at approximately twelve hour intervals.

<u>Method of administration</u>
Medovir tablets should be swallowed whole with a little water.

4.3. Contraindications
Medovir 400mg Tablets are contraindicated in patients known to be hypersensitive to the active substance, aciclovir or valaciclovir or to any of the excipients listed in section 6.1.

4.4. Special warnings and precautions for use
Use in patients with renal impairment and in elderly patients:
Aciclovir is eliminated by renal clearance, therefore the dose must be adjusted in patients with renal impairment. Elderly patients are likely to have reduced renal function and therefore the need for dose adjustment must be considered in this group of patients. Both elderly patients and patients with renal impairment are at increased risk of developing neurological side effects and should be closely monitored for evidence of these effects. In the reported cases, these reactions were generally reversible on discontinuation of treatment.

Hydration status: Care should be taken to maintain adequate hydration in patients receiving higher oral doses of aciclovir.

The result of a wide range of mutagenicity tests in vitro and in vivo indicate that aciclovir is unlikely to pose a genetic risk to man. Aciclovir was not found to be carcinogenic in long term studies in the rat and the mouse. Largely reversible adverse effects on spermatogenesis in association with overall toxicity in rats and dogs have been reported only at doses of aciclovir greatly in excess of those employed therapeutically. Two generation studies in mice did not reveal any effect of aciclovir on fertility. There is no experience of the effect of aciclovir on human female fertility. Aciclovir has been shown to have no definite effect upon sperm count, morphology or motility in man.

The data currently available from clinical studies is not sufficient to conclude that treatment with Medovir reduces the incidence of chickenpox-associated complications in immunocompetent patients.

4.5. Interactions with other medicinal products and other forms of interaction
No clinical significant interactions have been identified.
Aciclovir is eliminated primarily unchanged in urine via active renal tubular secretion. Any drugs administered concurrently that compete with this mechanism may increase aciclovir plasma concentrations. Probenecid and cimetidine increase the AUC of aciclovir by this mechanism and reduce aciclovir renal clearance. Similarly increases in plasma AUCs of aciclovir and of the inactive metabolite of mycophenolate mofetil, an immunosuppressant agent used in transplant patients have been shown when the drugs are co-administered. However no dosage adjustment is necessary because of the wide therapeutic index of aciclovir.

4.6. Pregnancy and lactation

A post-marketing aciclovir pregnancy registry has documented pregnancy outcomes in women exposed to any formulation of Medovir. The birth defects described amongst Medovir exposed subjects have not shown any uniqueness or consistent pattern to suggest a common cause.
Caution should however be exercised by balancing the potential benefits of treatment against any possible hazard.

Following oral administration of 200mg Medovir 5 times a day, aciclovir has been detected in breast milk at concentrations ranging from 0.6 to 4.1 times the corresponding plasma levels. These levels would potentially expose nursing infants to aciclovir dosages of up to 0.3mg/kg/day. Caution is therefore advised if aciclovir is to be administered to a nursing woman.
Limited human data show that the drug does not pass into breast milk following systemic administration.

4.7. Effects on ability to drive and use machines

There have been no studies to investigate the effect of aciclovir on driving performance or the ability to operate machinery. A detrimental effect on such activities cannot be predicted from the pharmacology of the active substance, but the adverse event profile should be borne in mind.

4.8. Undesirable effects

The frequency categories associated with the adverse events below are estimates. For most events, suitable data for estimating incidence were not available. In addition, adverse events may vary in their incidence depending on the indication.
The following convention has been used for the classification of undesirable effects in terms of frequency: very common ≥1/10, common ≥1/100 to <1/10, Uncommon ≥1/1000 to <1/100, Rare ≥1/10,000 to <1/1000, Very rare <1/10,000, Not known (cannot be estimated from the available data).

<u>Blood and lymphatic system disorders:</u>
Very rare: Anaemia, leukopenia, thrombocytopenia

<u>Immune system disorders:</u>
Rare: Anaphylaxis

<u>Psychiatric disorders and nervous system disorders:</u>
Common: Headache, dizziness.
Very rare: Agitation, confusion, tremor, ataxia, dysarthria, hallucinations, psychotic symptoms, convulsions, somnolence, encephalopathy, coma

The above psychiatric and nervous system events are generally reversible and usually reported in patients with renal impairment or with other predisposing factors.

<u>Respiratory, thoracic and mediastinal disorders:</u>
Rare: Dyspnoea

<u>Gastrointestinal disorders:</u>
Common: Nausea, vomiting, diarrhoea, abdominal pains

<u>Hepatobiliary disorders:</u>
Rare: Reversible rises in bilirubin and liver related enzymes.
Very rare: Hepatitis, jaundice

<u>Skin and subcutaneous tissue disorders:</u>
Common: Pruritus, rashes (including photosensitivity).
Uncommon: Urticaria. Accelerated diffuse hair loss. Accelerated diffuse hair loss has been associated with a wide variety of disease processes and medicines, the relationship of the event to aciclovir therapy is uncertain.
Rare: Angioedema.

<u>Renal and urinary disorders:</u>
Rare: Increases in blood urea and creatinine
Very rare: Acute renal failure, renal pain

General disorders and administration site conditions:
Common: Fatigue, fever

4.9. Overdose
Symptoms:
Aciclovir is only partly absorbed in the gastrointestinal tract. Patients have ingested overdoses of up to 20g aciclovir on a single occasion, usually without toxic effects.
Accidental, repeated overdoses of oral aciclovir over several days have been associated with gastrointestinal effects (such as nausea and vomiting) and neurological effects (headache and confusion).
Overdosage of intravenous aciclovir has resulted in elevations of serum creatinine, blood urea nitrogen and subsequent renal failure. Neurological effects including confusion, hallucinations, agitation, seizures and coma have been described in association with intravenous overdosage.

Management:
Patients should be observed closely for signs of toxicity. Haemodialysis significantly enhances the removal of aciclovir from the blood and may, therefore, be considered a management option in the event of symptomatic overdose.

5. PHARMACOLOGICAL PROPERTIES
5.1. Pharmacodynamic properties
Aciclovir is a synthetic purine nucleoside analogue which has *in vitro* and *in vivo* activity against human herpes virus, including *Herpes simplex* (type I and II) and *Varicella zoster*.
The inhibitory action of aciclovir is highly selective. Thymidine kinase enzyme in human cells does not use aciclovir as a substrate effectively, thus toxicity for mammalian cells is low. Virally produced thymidine kinase does use aciclovir, converting it to aciclovir monophosphate, which is further converted to diphosphate and then triphosphate by cellular enzymes. Aciclovir triphosphate interferes with viral DNA polymerase, and following incorporation in viral DNA inhibits DNA replication and causes chain termination.
Prolonged or repeated courses of aciclovir in severely immunocompromised individuals may result in the selection of virus strains with reduced sensitivity, which may not respond to continued aciclovir treatment. Most of the clinical isolates with reduced sensitivity have been relatively deficient in viral TK, however strains with altered viral TK or viral DNA polymerase have also been reported. *In vitro* exposure of HSV isolates to aciclovir can also lead to the emergence of less sensitive strains. The relationship between the *in vitro* determined sensitivity of the HSV isolates and clinical response to aciclovir therapy is not clear.

5.2. Pharmacokinetic properties
Aciclovir is only partially absorbed from the gut. Mean steady state peak plasma concentrations ($C^{ss}max$) following doses of 200mg administered four-hourly were 3.1microMol (0.7micrograms/ml) and equivalent trough plasma levels ($C^{ss}min$) were 1.8microMol (0.4micrograms/ml). Corresponding $C^{ss}max$ levels following doses of 400mg and 800mg administered four-hourly were 5.3microMol (1.2micrograms/ml) and 8microMol (1.8micrograms/ml) respectively and equivalent $C^{ss}min$ levels were 2.7microMol (0.6micrograms/ml) and 4microMol (0.9micrograms/ml).
In adults the terminal plasma half-life of aciclovir after administrations of intravenous aciclovir is about 2.9 hours. Most of the drug is excreted unchanged by the kidney. Renal clearance of aciclovir is substantially greater than creatinine clearance, indicating that tubular secretion, in addition to glomerular filtration contributes to the renal elimination of the drug. 9-carboxymethoxymethylguanine is the only significant metabolite of aciclovir, and accounts for approximately 10 - 15% of the administered dose recovered from the urine. When aciclovir is given one hour after 1 gram of probenecid the terminal half-life and the area under the plasma concentration time curve is extended by 18% and 40% respectively.
In adults, mean steady state peak plasma concentrations ($C^{ss}max$) following a one hour infusion of 2.5 mg/kg, 5 mg/kg and 10mg/kg were 22.7microMol (5.1micrograms/ml), 43.6microMol (9.8micrograms/ml) and 92microMol (20.7micrograms/ml), respectively. The corresponding trough levels ($C^{ss}min$) 7 hours later were 2.2microMol (0.5micrograms/ml), 3.1microMol (0.7micrograms/ml), and 10.2microMol (2.3micrograms/ml), respectively.
In children over 1 year of age similar mean peak ($C^{ss}max$) and trough ($C^{ss}min$) levels were observed when a dose of 250mg/m^2 was substituted for 5mg/kg and a dose of 500mg/m^2 was substituted for 10mg/kg. In neonates and young infants (0 to 3 months of age) treated with doses of 10mg/kg administered by infusion over a one-hour period every 8 hours the $C^{ss}max$ was found to be 61.2microMol (13.8micrograms/ml) and $C^{ss}min$ to be 10.1microMol (2.3micrograms/ml). The terminal plasma half-life in these patients was 3.8 hours.
In the elderly, total body clearance falls with increasing age associated with decreases in creatinine clearance although there is little change in the terminal plasma half-life.
In patients with chronic renal failure the mean terminal half-life was found to be 19.5 hours. The mean aciclovir half-life during haemodialysis was 5.7 hours. Plasma aciclovir levels dropped approximately 60% during dialysis.

Cerebrospinal fluid levels are approximately 50% of corresponding plasma levels. Plasma protein binding is relatively low (9 to 33%) and drug interactions involving binding site displacement are not anticipated.

6. PHARMACEUTICAL PARTICULARS

6.1. List of excipients

Lactose spray dried
Microcrystalline cellulose
Croscarmellose sodium
Magnesium stearate
Erythrosine aluminium lake (E127)

6.2. Incompatibilities

No known incompatibilities.

6.3. Shelf life

60 months.

6.4. Special precautions for storage

Store below 30°C in the original packaging

6.5. Nature and contents of container

Combination of polyvinylchloride and aluminium blisters in a carton with the patient information leaflet. Cartons of 10, and 100 tablets are available. Not all pack sizes may be marketed.

6.6. Special precautions for disposal

No special requirements for disposal.

7. PRODUCT REGISTRATION HOLDER

KOMEDIC SDN BHD, No. 4 Jalan PJS 11/14, Bandar Sunway, 46150 Petaling Jaya Selangor, MALAYSIA

8. MANUFACTURER

MEDOCHEMIE (FAR EAST) LTD., (Oral Facility), No. 40, VSIP II Street 6, Vietnam-Singapore Industrial Park II, Binh Duong Industry-Service-Urban Complex, Hoa Phu ward, Thu Dau Mot city, Binh Duong province, Vietnam

9. PRODUCT REGISTRATION NUMBER

MAL09062118AZ

10. DATE OF REVISION OF THE TEXT

October 2022